

Omar Gamal

Backend & Infrastructure Engineer | Go, Kubernetes, Distributed Systems

Cairo, Egypt (UTC+2/+3) | +20 106 444 4598 | omargamal.dev@outlook.com

github.com/Omar0Gamal | linkedin.com/in/omar-gamal-091044168 | portfolio-omar0gamals-projects.vercel.app

Remote | Fluent English | Available full-time | Can overlap with EU/US hours

Summary

Backend engineer who designed the architecture of a 14-service enterprise management system (Kubernetes, Helm, Terraform) under contract, and open-sourced Cartwright, a saga-based distributed checkout system (Go, C#, Python) demonstrating the same patterns publicly. Comfortable owning a service from API design through CI/CD and deployment. Final-year Computer Science student.

Education

Capital University

Expected 2027

B.Sc. Computer Science | GPA: 3.92 / 4.0 (Dean's List)

Cairo, Egypt

Technical Skills

Languages: Go, C++, Java, Python, C#, SQL

Backend & APIs: REST, OpenAPI, gRPC, Protocol Buffers, Gin, sqlc, ASP.NET, microservice architecture, distributed systems, role-based access control, multi-tenancy

Messaging & Data: NATS JetStream, PostgreSQL, Redis, SQLite, MongoDB, MySQL

Infrastructure: Docker, Kubernetes, Helm, Terraform, Linux, Caddy, Git, GitHub Actions, CI/CD

Observability: Prometheus, Grafana, Loki, Promtail, request tracing

Systems: Multithreading (lock striping), idempotency & retry design, CMake

Experience

Project Lead (Contract)

Sep 2025 – Nov 2025

Software company (name withheld, under NDA)

- Designed a 14-service microservice architecture (ERP, CRM, document management) and supervised initial development for a team of 6 (Go, Python, React); the project ended when the company's only client withdrew.
- Wrote all Protocol Buffers definitions for internal service communication, documented cross-service call order and failure recovery, and wrote the Terraform and Helm chart to deploy on Kubernetes on Hetzner.

Backend Developer & Server Administrator

2022 – Mar 2026

Attila Gaming (Remote)

- Built matchmaking, game-instance spawning/lifecycle management, and anti-cheat features across 15+ Java/Spigot plugins for a Minecraft server with 500+ concurrent players.
- Administer the Linux servers (CI/CD, containerized services, automated backups); reduced crashes by 60% and improved response times by 40%.

Freelance Software Developer

2018 – 2022

Self-Employed

Remote

- Delivered 20+ projects for international clients, from Discord bots and small websites (2018) to web applications, APIs, and AI-powered chatbot integrations.

Projects

Cartwright | Go, C#/ASP.NET, Python, gRPC, NATS, K8s

github.com/Omar0Gamal/cartwright

- Saga-based distributed order-checkout system: Go orchestrator, C#/ASP.NET billing service, Python notifier over gRPC/Protocol Buffers; compensating transactions and a transactional outbox to NATS JetStream.
- Idempotency keys, backoff retries, and lease-based recovery; a scripted chaos test kills the billing service mid-request and confirms no double charge on recovery. Deployed via Helm to k3s provisioned with Terraform; CI across all three languages.

AxiomGraph | C++17, Python, SQLite, CMake

github.com/Omar0Gamal/AxiomGraph

- Hybrid graph-vector database in C++17: adjacency-list graph (CSR-consolidated) plus vector KNN search (hnswlib CPU / NVIDIA cuVS GPU); lock-striped concurrent mutations; nanobind Python SDK; wheels released on GitHub.

Open Source

llama.cpp – RPC server bounds-check fix | C++

github.com/ggml-org/llama.cpp/pull/26933 (open)

- Wrote a fix for two reported out-of-bounds memory bugs in llama.cpp's RPC server (a GET_ROWS read and a SET_ROWS write): validates tensor shapes before graph execution and rejects malformed requests. Pull request open, awaiting maintainer review.